

Adaptive Valve Profile for Postural ICP Regulation

Executive Technology Brief

Domain: Fluid Mechanics + Adaptive Valve Design

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: ICP excursions during postural changes cause patient symptoms

Mechanism: Valve whose opening pressure profile adapts to ICP trend via trend-feedback control

Architecture: CSF shunt valve whose opening pressure profile adapts to ICP trend rather than fixed cracking pressure; reduces postural pressure excursions

Governing Model: Variable-area hydraulic orifice with trend-feedback control. The valve is modeled as a non-linear orifice whose area A is a function of measured pressure and its time-derivative.

Key Equations:

$Q = C_d * A * \sqrt{2 * dP / \rho}$ [turbulent/orifice flow]

$Q = (\pi * r^4 * dP) / (8 * \eta * L)$ [laminar flow through circular conduit]

$Re = (\rho * v * D) / \eta$ [laminar if $Re < 2300$]

$A_{actuator}(P, dP/dt) = A_0 + k_p * (P - P_0) + k_d * dP/dt$ [proposed controller law, MODELLED]

Kill Condition: Adaptive response time exceeds postural transient duration (cannot respond within seconds)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.